# ON THE NON-EXISTENCE OF NON-TRIVIAL COLLATZ CYCLES: A CONDITIONAL FORMAL PROOF IN LEAN 4 WITH DOCUMENTED STRUCTURAL OBSTRUCTIONS

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ABSTRACT. We address the no-non-trivial-cycle disjunct of the Collatz conjecture. The central result, formalized in Lean 4 with Mathlib v4.27.0, is a conditional non-existence theorem declared parametrically with three structural hypotheses: `BakerSeparation` (after Baker 1966), `BarinaVerification` (after the 2025 computational verification that every positive integer below `2⁷¹` reaches 1), and `ProductBound–Threshold`, a project-derived cycle-complexity bound made fully explicit in §4.3. The Lean formalization is machine-verified with kernel-3 axiom profile, no user axioms and no `sorry`, and is reproducible via `reproduce.sh` against an explicit `expected_axioms.md` baseline.

We complement the theorem with two impossibility lemmas (δ8, δ8') showing that no uniform algebraic refinement of the Product Bound derivation can eliminate the third hypothesis within the Baker + continued-fraction framework; a literature mapping (δ9) documenting the absence of any peer-reviewed deterministic upper bound on cycle length in the 1977-2026 results we surveyed; and an alternative disjunctive framing (δ7) that connects the conditional theorem to Hercher's 2023 lower bound. The paper is intended as a stable substrate for future contributions resolving the structural conditionality.

Keywords. Collatz conjecture, 3x+1 problem, Collatz cycles, linear forms in logarithms, Baker's theorem, continued fractions, irrationality measure, formal verification, Lean 4, Mathlib.

MSC 2020. 11B83 (Special sequences), 11J81 (Transcendence; general theory of irrational numbers), 11J86 (Linear forms in logarithms), 11Y55 (Calculation of integer sequences), 37P99 (Arithmetic and non-Archimedean dynamical systems), 68V15 (Theorem proving and proof assistants).

## 1. Introduction

1.1. The Collatz conjecture and the cycle problem. For each positive integer $n$, the Collatz map $T : \mathbb{N} \to \mathbb{N}$ is defined by $T(n) = n/2$ if $n$ is even and $T(n) = (3n+1)/2$ if $n$ is odd. The Collatz conjecture asserts that, for every starting value, iterating $T$ eventually reaches 1. Equivalently, it makes two negative claims: (i) no trajectory diverges to infinity, and (ii) no non-trivial periodic orbit exists — where *non-trivial* means distinct from the fixed orbit $1 \to 1$. The present paper addresses (ii) only; the divergence half of the conjecture is outside our scope.

We find it convenient to work with the *odd iterate* $T_{\text{odd}}$, which compresses runs of halvings and advances the index by one only on a $(3n+1)$ step. A pair $(n, k)$ with $n$ odd, $k \geq 1$, and $T^k_{\text{odd}}(n) = n$ is called an odd Collatz cycle of length $k$. The trivial cycle $(1, 2)$ gives the trivial period $1 \to 1$. Our central question is whether any pair $(n, k)$ with $n > 1$ is an odd Collatz cycle.

1.2. Motivation and scope. A natural starting point is the conditional reduction of the cycle problem to two published external results: Baker's 1966 theorem on linear forms in logarithms of algebraic numbers, and Barina's 2025 computational verification that every integer below $2^{71}$ ultimately reaches 1. In early 2026 we set out to remove the second of these, or at least to reduce the verification range substantially, by promoting an intermediate structural hypothesis of our Lean formalization to a Lean theorem. A deliberately exhaustive exploration of the available methodology returned a single structural verdict: every product-bound approach known to the literature is limited to cycle length at most of order $10^{10}$, and any successful bridge from Baker's theorem through the continued-fraction theory of $\log_2 3$ to the cycle-length bound we require would have to surmount an obstruction we formulate here as the Product-Bound Impossibility Lemma (§5, Lemma 5.1).

The obstruction is not a gap in our own effort alone; it is a consequence of the irrationality measure of $\log_2 3$ and is shared by every argument in the same family. It is also worth documenting that the obstruction cuts off not just our approach but, as we detail in §6, the closest 2026 preprints as well. The present paper therefore (a) states the conditional theorem cleanly, (b) records the three structural hypotheses that it rests on, and (c) explains the obstruction and the state of the art that make the third hypothesis necessary. The third hypothesis is named ProductBoundThreshold and is made explicit in §4.3.

1.3. Contributions. The paper contains five originals, each with a rigorous statement later in the text. To make cross-section navigation easier we use five short labels — $\delta 7$, $\delta 8$, $\delta 8'$, $\delta 9$, and $6\alpha$ — each of which is defined verbatim in the corresponding bullet below. These labels are *presentational shorthand only* (they have no mathematical content beyond what the bulleted text states) and are reused in the §10 conclusion to keep the contributions traceable.

- $\delta 7$ (alternative framing, §7). A logically equivalent presentational restatement of the central theorem as the disjunction "$k \leq 1322$ or $n > 2^{71}$". The reformulation is not an independent theorem (it is equivalent to Theorem 3.1 under the same three hypotheses); we list it as a contribution because the disjunctive form makes the bridge to Hercher's (2023) lower-bound regime transparent at a glance, even though it does not close the cycle problem.
- $\delta 8$ (Product-Bound Impossibility Lemma, §5.1). A meta-mathematical lemma: any argument that bounds a hypothetical cycle minimum $m$ by a polynomial-over-cycle-length $F(k)$ with $F(k) < 2^{71}$ uniformly in $k$ forces $\log_2 3$ to be rationally approximable within a constant factor, contradicting its irrationality. The lemma explains why no refinement of Baker's effective exponent within the standard framework can discharge our third hypothesis.
- $\delta 8'$ (extended impossibility, §5.2). A corollary of $\delta 8$ extending the obstruction to the family of bound schemes obtained by composing Baker-type inequalities with Steiner's cycle equation

and the continued-fraction convergents $p_j/q_j$ of $\log_2 3$. Numerical corroboration is given window by window, connecting with Khinchin's best-second-kind characterization of convergents.

- $\delta 9$ (state-of-the-art mapping, §6). A typology of the 1977-2026 Collatz cycle literature distinguishing lower bounds on cycle length, structural-class eliminations (Steiner 1977 circuits; Knight 2026 high cycles), probabilistic density results (Terras 1976; Tao 2019/2022), and recent reformulation attempts (Santana 2026; Dhiman-Pandey 2026; Rozier-Terracol 2026). The typology makes explicit, to the best of our literature review, that no published result supplies the deterministic upper bound on cycle length that our third hypothesis encodes.

- 6α (formal verification, §8). The central theorem and its immediate dependencies are formalized in Lean 4 under Mathlib v4.27.0, with zero user-declared axioms, zero `sorry`, and an axiom profile whose central chain consists of the three Mathlib kernel axioms (`propext`, `Classical.choice`, `Quot.sound`). Arithmetic gap constants used in auxiliary roles are isolated from the central chain through a structural parameter, so that the two Lean compiler axioms required by `native_decide` remain outside it at the present formalization state.

1.4. Relation to the 2026 literature. Four recent contributions warrant explicit positioning relative to our conditional theorem.

- Santana (2026, arXiv:2601.03297v4) proposes a topological and ergodic reformulation of the cycle problem. We directly consulted the full preprint. The argument for Theorem B (finiteness) relies on an unjustified boundedness assumption for a sequence of integrals over atomic invariant measures, and Lemma 16, which would extend finiteness to uniqueness, is labeled by the author (Remark 17) as "an alternative approach [...] rather than a proof". The framework therefore does not discharge our third hypothesis, although it is complementary (§6.4).

- Knight (2026, *Discrete Math.* 349(3), 114812) proves that Collatz "high cycles" do not exist. This is a restricted-class elimination in the tradition of Steiner's

(1977) circuits result; it does not extend, without substantial further work, to general parity patterns (§6.2). Access to the full text was blocked at the time of writing; we therefore rely on the abstract-level formulation validated by our upstream survey and flag the claim as such.

- Dhiman-Pandey (2026, arXiv:2601.12772) prove, by a 2-adic "ghost-cycle" construction, that cycle equations cannot be characterized in Presburger arithmetic. This is methodologically orthogonal to our Baker-plus-continued-fractions approach: the two results rule out different proof families (§5.3).

- Rozier-Terracol (2025, arXiv:2502.00948, to appear *Discrete Math.* 2026) enumerate so-called *paradoxical sequences* and use Rhin's effective irrationality bound heuristically. Their finiteness statement for paradoxical sequences is independent of our third hypothesis (§6.4).

1.5. Paper organization. Section 2 fixes notation and restates the three hypothesis-structures as they appear in the Lean 4 formalization. Section 3 gives the central conditional theorem. Section 4 discusses the mathematical origin and the published support of each of the three hypotheses. Section 5 proves the Product-Bound Impossibility Lemma (δ8) and its extension (δ8'). Section 6 presents the literature mapping (δ9). Section 7 gives the disjunction framing (δ7). Section 8 describes the Lean 4 formalization (6α), including its axiom profile and its reproducibility contract. Section 9 lists open problems and connects with ongoing speculative tracks not integrated here. Section 10 concludes. Section 11 is the reference list.

## 2. Framework and notation

2.1. The Collatz map, the odd iterate, and odd cycles. The Collatz map `T: ℕ_{≥1} → ℕ_{≥1}` is defined by `T(n) = n/2` when `n` is even, and `T(n) = (3n+1)/2` when `n` is odd. The odd iterate `T_odd` compresses the halvings: starting from an odd integer, `T_odd(n)` is the unique odd integer on the Collatz trajectory of `n` reached after exactly one `3n+1/2` step followed by all consecutive halvings. In the Lean 4 formalization accompanying this paper `T_odd` is written `syracuseNext`, and its iterate is the function `nSeq: ℕ → ℕ → ℕ` of `ProjetCollatz/SyracuseDefs.lean` line 213, defined by `nSeq start 0 = start` and `nSeq start (k+1) = syracuseNext (nSeq start k)`.

Definition 2.1 (Odd Collatz cycle). A pair $(n, k)$ of natural numbers is an *odd Collatz cycle* if $n > 1$, $n$ is odd, $k \geq 1$, and `nSeq` $n\,k = n$.

This corresponds to the predicate `IsOddCycle` of `ProjetCollatz/Phase50CycleEquation.lean` lines 27-28:

```
def IsOddCycle (n: ℕ) (k: ℕ): Prop :=
  n > 1 ∧ n % 2 = 1 ∧ k ≥ 1 ∧ nSeq n k = n
```

The exclusion `n > 1` removes the trivial fixed orbit at 1; every reference below to "cycle" means *odd Collatz cycle* in this sense. The cycle problem is the negative claim: no such `(n, k)` exists.

2.2. Parity vectors, the sum of halving exponents, and Steiner's identity. Fix an odd Collatz cycle $(n, k)$. Write $n_i$ := `nSeq` $n\,i$ for $0 \leq i \leq k$, so that $n_0 = n_k = n$ and each $n_i$ is odd. Between $n_i$ and $n_{i+1}$ there is exactly one $(3n+1)/2$ step followed by some number $s_{i+1} \geq 0$ of halvings, called the $i$-th *parity exponent*. The list $(s_1, \ldots, s_k)$ is the *parity vector* of the cycle, and $s := s_1 + \cdots + s_k$ its *sum of halving exponents*.

Collecting all $k$ odd steps yields Steiner's cycle identity:

$$n \cdot (2^s - 3^k) \;=\; C_{n,k,s},$$

where

$$C_{n,k,s} \;=\; \sum_{i=1}^{k} 3^{k-i}\, 2^{s_1+\cdots+s_{i-1}}$$

is a strictly positive integer (the "corrective sum"). The identity is the discrete analogue of log-linearity of the Collatz dynamics and is the conventional entry point to Baker-style arguments. Its Lean formalization is in `ProjetCollatz/Phase52SteinerEquation.lean` (`corrSum`, `steiner_eq`, `steiner_cycle_eq`, `corrSum_pos_of_cycle`, lines 101-150).

A direct consequence of Steiner's identity is that $2^s > 3^k$ for any hypothetical non-trivial cycle (else the right-hand side would be non-positive), so $s = \lceil k \cdot \log_2 3 \rceil$ whenever the cycle is compatible with the ambient inequality framework. This is the bridge to continued-fraction approximation theory (§7).

2.3. The three structural hypotheses. Our central theorem depends on three `structure` fields, all declared in `ProjetCollatz/Phase58PorteDeuxFinal.lean`.[1]

*`BakerSeparation` (Phase58, lines 67-69).*

```
structure BakerSeparation where
  separation : ∀ (s k : ℕ), s ≥ 1 → k ≥ 2 → 2^s > 3^k →
    (2^s - 3^k) * k^6 ≥ 3^k
```

[1]Module names follow the project's bilingual French/English convention (e.g., "PorteDeux" = "Door Two"); mathematical content is unaffected.

This is the effective version of Baker's 1966 theorem specialized to the cycle problem, at irrationality-measure exponent $\mu = 6$ — a strictly weaker constant than Salikhov's (2007) sharper bound $\mu(\ln 3) \leq 5.125$ (refining Rhin's 1987 linear-form bound), but sufficient for our derivation and straightforward to state. See §4.1.

*`BarinaVerification` (Phase58, lines 80-81).*

```
structure BarinaVerification where
  convergence: ∀ (n: ℕ), n > 0 → n < 2^71 → reaches_one n
```

Barina's 2025 computational verification that every positive integer below $2^{71}$ ultimately reaches 1. See §4.2.

*`ProductBoundThreshold` (Phase58, lines 296-297).*

```
structure ProductBoundThreshold where
  cycle_length_bound: ∀ (n k: ℕ), IsOddCycle n k → k ≤ 982
```

This is the third hypothesis. Its origin is derived, not taken from a single published paper: combining the Product Bound lemma of `ProjetCollatz/Phase56*.lean`, which states $m \leq (k^7 + k)/3$ for a cycle minimum $m$ (a consequence of Baker's separation inequality), with Barina's limit $2^{71}$ yields the optimal threshold $k \leq 1322$, certified by the arithmetic inequality

$$(1322^7 + 1322)/3 \ < \ 2^{71} \ < \ (1323^7 + 1323)/3.$$

The structure parameter is fixed at the conservative value $k \leq 982$ (an $\approx 8\times$ safety margin below the optimal threshold), which produces the simpler arithmetic certificate $(982^7 + 982)/3 < 2^{71}$. Both bounds (the optimal 1322 and the conservative 982) are mechanically verified: the former by `product_bound_fits_barina_1322` and the latter by `k982_bound`, both in Phase56-58 via `native_decide`. The derivation is honest and transparent, but the promotion of this hypothesis to a theorem is structurally blocked: this is the content of §5. See also `ProjetCollatz/HYPOTHESES.md` in the accompanying repository for the full derivation chain. Throughout this paper, $k$ denotes the number of odd steps in the cycle, and $m$ denotes the cycle minimum integer (smallest odd element of the cycle); this $m$ notation appears in §3.2 step 1 and corresponds to the Phase56 lemma `cycle_has_min`. Note that Simons-de Weger (2005) use $m$ for a distinct quantity (number of local minima in the cycle); their convention is not adopted here, and their bound $m > 68$ is reproduced verbatim in §6.1 with explicit author attribution.

2.4. The central theorem (forward pointer). Theorem (informal statement, see §3 for the formal version). Assume `BakerSeparation`, `BarinaVerification`, and `ProductBoundThreshold`. Then there is no odd Collatz cycle in the sense of Definition 2.1.

In the Lean formalization this is `ProjetCollatz.no_nontrivial_cycle_final` in `ProjetCollatz/Phase58PorteDeuxFinal.lean` line 339. Its proof is discharged in six elementary steps, recorded verbatim in §3 and documented with line numbers in §8.

2.5. Notation conventions.

- $\log_2 3$ denotes the real number `Real.logb 2 3` of Mathlib 4.
- Continued-fraction convergents of $\log_2 3$ are written $p_n/q_n$ with $q_n \in \mathbb{N}_{\geq 1}$ the denominator; these correspond to the Lean-side notation `q_n` of `ProjetCollatz/Phase61CFConvergents.lean`.
- The six *windows* used in §7 are defined by $W_n := [q_n, q_{n+1})$ for $n \in \{8, 9, 10, 11, 12, 13\}$, the predicate `InWindow n k ↔ q_n ≤ k < q_{n+1}` being the Lean abbreviation of `ProjetCollatz/Phase61CFConvergents.lean`.
- Numerical values: $q_8 = 665$, $q_9 = 15{,}601$, $q_{10} = 31{,}867$, $q_{11} = 79{,}335$, $q_{12} = 111{,}202$, $q_{13} = 190{,}537$, $q_{14} = 10{,}590{,}737$ (Phase59 `cf_nbound_*` constants).

- `reaches_one` is the Lean abbreviation for "the Collatz trajectory starting at $n$ eventually hits 1"; used in the `BarinaVerification` field and in the contradiction-on-cycle proof.

All other symbols introduced later are local to their section.

2.6. Summary of the three structural hypotheses. The three structural hypotheses underlying the conditional theorem of §3 are summarized below. Their mathematical and bibliographic context is detailed in §4; their formal Lean signatures are quoted verbatim in §2.3.

| Symbol | Origin | Status | Mathematical content |
|---|---|---|---|
| `BakerSeparation` | Baker (1966), Rhin (1987), Matveev (2000) | Published; not Lean-formalized | $(2^s - 3^k)\,k^6 \geq 3^k$ for $s \geq 1,\ k \geq 2,\ 2^s > 3^k$ |
| `BarinaVerification` | Barina (2025) | Computational; not Lean-formalized | $\forall n > 0.\ n < 2^{71} \Rightarrow$ `reaches_one`$(n)$ |
| `ProductBoundThreshold` | Project-derived (§4.3) | Hypothesis; obstruction in §5 | $\forall n, k.$ `IsOddCycle`$(n, k) \Rightarrow k \leq 982$ |

The three hypotheses are exhibited as Lean `structure` parameters of `no_nontrivial_cycle_final` (§3.1, §8.1); the conditional theorem is independent of the *content* of these structures, depending only on their declared signatures.

2.7. Notation index. For ease of reference, the principal symbols introduced in this section and used recurrently throughout the paper are collected here.

| Symbol | Meaning | First occurrence |
|---|---|---|
| $T$ | Collatz map $T : \mathbb{N}_{\geq 1} \to \mathbb{N}_{\geq 1}$ | §1.1 |
| $T_{\text{odd}}$ | Odd iterate (compresses halvings) | §1.1 |
| `nSeq` | Lean encoding of $T_{\text{odd}}^k$ | §2.1 |
| `IsOddCycle` | Lean predicate $(n, k) \mapsto$ cycle | §2.1, Definition 2.1 |
| $n, k$ | Cycle starting point and length | §2.1 |
| $m$ | Cycle minimum integer | §2.3.3, §3.2 |
| $s, s_i$ | Halving exponents and parity vector | §2.2 |
| $C_{n,k,s}$ | Steiner corrective sum | §2.2 |
| $\log_2 3$ | Real number, irrationality measure $\leq 5.125$ | §2.5 |
| $p_n/q_n$ | Continued-fraction convergents of $\log_2 3$ | §2.5 |
| $W_n$ | Continued-fraction window $[q_n, q_{n+1})$ | §2.5 |
| `BakerSeparation` | First structural hypothesis (Baker 1966) | §2.3.1, §4.1 |
| `BarinaVerification` | Second structural hypothesis (Barina 2025) | §2.3.2, §4.2 |

| Symbol | Meaning | First occurrence |
|---|---|---|
| `ProductBoundThreshold` | Third structural hypothesis (project-derived) | §2.3.3, §4.3 |
| `DerivedLargeKBound` | CF-side equivalent of `ProductBoundThreshold` | §8.1, Appendix D |
| $\delta 7, \delta 8, \delta 8', \delta 9$ | Paper contributions (informal codes) | §1.3 |
| $6\alpha$ | Paper contribution: formal-verification artifact | §1.3 |
| $\Phi_s, \Psi_s, \psi_s$ | Structural-excess functions (supplementary) | §8.5, §9.7, Appendix A |

## 3. Main theorem: conditional proof

3.1. Statement. We formalize the following conditional result in Lean 4 (Mathlib v4.27.0).

Theorem 3.1 (Conditional non-existence of non-trivial Collatz cycles). *Let `baker : BakerSeparation`, `barina : BarinaVerification`, and `sdw : ProductBoundThreshold`. Then for every pair of natural numbers n, k, the predicate `IsOddCycle n k` entails `False`.*

The formal Lean statement (verbatim from the project, file `ProjetCollatz/Phase58PorteDeuxFinal.lean`, line 339) reads:

```
theorem no_nontrivial_cycle_final
    (baker : BakerSeparation) (barina : BarinaVerification)
    (sdw : ProductBoundThreshold)
    (n k : ℕ) (hcyc : IsOddCycle n k) : False
```

See §2.3 for the Lean `structure` fields underlying the three hypothesis parameters, §4 for the mathematical origin of each hypothesis, and §8.1 for the full aliases listing the six equivalent packagings of Theorem 3.1 exported by the repository.

Note on hypothesis redundancy. The third hypothesis `ProductBoundThreshold` is *interderivable* with the continued-fraction-side hypothesis `DerivedLargeKBound` (§8.1) in the presence of `BakerSeparation` and `BarinaVerification`: the bridge lemma `sdw_from_cf` (Phase59 line 197) constructs the former from the latter. The two formulations are therefore logically equivalent, and the canonical theorem can be equivalently stated under either of the two three-hypothesis sets `(BakerSeparation, BarinaVerification, ProductBoundThreshold)` or `(BakerSeparation, BarinaVerification, DerivedLargeKBound)`. We adopt the former in §3.1 because `ProductBoundThreshold` is the directly cycle-length-bounded statement (and thus the most natural target for elimination-as-theorem); we adopt the latter in §8 (Phase59 packaging) because it is the formulation that the continued-fraction infrastructure actually produces.

3.2. Proof chain.

1. Extract the cycle minimum $m$ via `cycle_has_min` (Phase56, proved).
2. Apply `cycle_min_bound_nat` (Phase56, proved, uses Baker): $m \leq (k^7 + k)/3$.

3. Use `ProductBoundThreshold.cycle_length_bound`: $k \le 982$. This is the structure parameter (a conservative bound with $\approx 8\times$ safety margin); the corresponding optimal Lean theorem `no_cycle_k_le_1322` (cf. §4.3 and §8.1) covers the larger range $k \le 1322$.
4. Apply `k982_bound` (Phase56, `native_decide`): $(982^7 + 982)/3 < 2^{71}$.
5. Hence $m < 2^{71}$. Combined with $m > 0$ (from `IsOddCycle`), Barina gives `reaches_one m`.
6. `cycle_prevents_reaching_one` (Phase50, proved) yields a contradiction.

3.3. Axiom profile. Note on perspective (parametric primary). `no_nontrivial_cycle_final` is declared parametrically with `sdw : ProductBoundThreshold` as an explicit parameter. The proof uses `sdw.cycle_length_bound` directly, without unfolding any specific witness. Therefore, the theorem-as-declared has axiom profile `[propext, Classical.choice, Quot.sound]` (kernel-3), as machine-verified by `reproduce.sh` against the `expected_axioms.md` baseline (see §8.6 / §8.7).

The alternative five-axiom reading below corresponds to the fully-instantiated proof term obtained when a caller supplies the concrete witness `k982_bound` (whose proof uses `native_decide`, introducing `Lean.ofReduceBool` and `Lean.trustCompiler`). This perspective is helpful for readers reconstructing the end-to-end argument but does not modify the axiom profile of the parametric theorem itself.

Instantiated profile. `propext`, `Classical.choice`, `Quot.sound` (kernel-3) + `Lean.ofReduceBool`, `Lean.trustCompiler` (from `native_decide` on `k982_bound`). All documented in `expected_axioms.md`.

## 4. The three structural hypotheses

The conditional theorem of §3 depends on three hypothesis-structures declared in `Projet-Collatz/Phase58PorteDeuxFinal.lean`. The Lean signatures are shown verbatim in §2.3 (`BakerSeparation` lines 67-69, `BarinaVerification` lines 80-81, `ProductBoundThreshold` lines 296-297); this section provides the mathematical and bibliographic context for each.

### 4.1. BakerSeparation (Baker 1966).

```
structure BakerSeparation where
  separation : ∀ (s k : ℕ), s ≥ 1 → k ≥ 2 → 2^s > 3^k →
    (2^s - 3^k) * k^6 ≥ 3^k
```

Source. Baker, A. (1966), *Linear forms in the logarithms of algebraic numbers I*, *Mathematika* 13, pp. 204-216. Refined by Rhin (1987) and Matveev (2000). Fields Medal 1970.

Scope. It is standard in the Collatz cycle literature to adopt Baker as an external hypothesis; Steiner (1977), Simons-de Weger (2005), and Hercher (2023) all use variants. Formalizing Baker's theorem in Lean would require approximately 10{,}000 lines of transcendence theory (Feldman-Nesterenko-Shorey-Tijdeman framework). The specific Baker-type inequality $(2^s - 3^k) \cdot k^{\mu} \ge 3^k$ follows from the standard reduction documented in Steiner (1977, §2) and Simons-de Weger (2005, Lemma 2.1). The hypothesis as stated absorbs Baker's effective constant $C$ into the unit prefactor — that is, it asserts the inequality at the idealized $C = 1$. For the literal Baker / Rhin / Matveev constants this absorption is valid asymptotically (for $k$ large enough that $C \cdot k^c$ exceeds 1 and the polynomial regime dominates), and no concrete lower bound on the relevant $k_0$ is required for the conditional theorem of §3 since the small-$k$ regime is closed independently by Hercher (2023) and by `no_cycle_k_le_1322` (Phase58).

Effective constant. The formalization uses the irrationality-measure exponent $\mu = 6$, strictly weaker than the sharpest known bound on $\mu(\ln 3)$. Rhin's (1987) linear-form bound for $|p \cdot \ln 2 + q \cdot \ln 3|$ yields the irrationality measure $\mu(\ln 3) \le 8.616$; Salikhov (2007) sharpens this to $\mu(\ln 3) \le 5.125$, and the linear-independence measure result of Wu (2003) corresponds to a slightly sharper formulation ($\mu(\ln 3) \le 5.117$).

Throughout this paper, three closely related but distinct quantities appear and we adopt the following convention. The *exponent* $c$ of an effective lower bound of the form $|p \cdot \ln 2 - q \cdot \ln 3| \geq C/H^c$ (with $H = \max(|p|, |q|)$) is invariant under the rational rescaling $\log_2 3 = \ln 3/\ln 2$; that is, the irrationality-measure exponents of $\ln 3$ and of $\log_2 3$ coincide. The *effective constant* $C$, however, is not invariant: rewriting the linear form as $|p - q \cdot \log_2 3| \cdot \ln 2 = |p \cdot \ln 2 - q \cdot \ln 3|$ introduces a factor $(\ln 2)^{-1}$ — and, in higher-power forms, $(\ln 2)^{-c}$ — when transferring between ln-bounds and $\log_2$-bounds. We track this factor explicitly in §5.1 where it matters. The structure `BakerSeparation` exposes only the exponent $\mu = 6$, not the multiplicative constant; this is by design — the $\mu = 6$ exponent alone suffices for the Product-Bound derivation chain.

4.2. BarinaVerification (Barina 2025).

```
structure BarinaVerification where
  convergence: ∀ (n: ℕ), n > 0 → n < 2^71 → reaches_one n
```

Source. Barina, D. (2025), *Improved verification limit for the convergence of the Collatz conjecture*, *Journal of Supercomputing* 81:810. DOI: 10.1007/s11227-025-07337-0.

Actual limit. $2075 \cdot 2^{60} \approx 2^{71.02}$; the bound $n < 2^{71}$ used in the formalization is slightly conservative.

Reproducibility. Barina's code is open-source; the computational verification relies on modular sieving and is reproducible (though it requires ~months of CPU time).

4.3. ProductBoundThreshold (project-derived, documented).

```
structure ProductBoundThreshold where
  cycle_length_bound: ∀ (n k: ℕ), IsOddCycle n k → k ≤ 982
```

Origin. This hypothesis is not a direct result from any single published paper (see `ProjetCollatz/HYPOTHESES.md` in the accompanying repository). It is a *cycle-complexity bound* whose explicit threshold $k \leq 982$ derives from:

1. The Product Bound lemma (`ProjetCollatz/Phase56*.lean`, proved algebraically from Baker plus Bernoulli): the cycle minimum $m$ satisfies $m \leq (k^7 + k)/3$.
2. Barina's verification limit $2^{71}$.
3. The arithmetic fact $(982^7 + 982)/3 < 2^{71}$, verified by `native_decide` in `k982_bound` (`Phase56Bloc18Complete.lean` line 249).

Status. For hypothetical cycles, $k \leq 982$ is vacuously true assuming the Collatz no-cycle conjecture. It is *stronger* than Hercher's (2023) lower bound $K > 1.375 \cdot 10^{11}$, but this apparent contradiction is resolved because both claims are vacuous on the (conjectured) empty set of non-trivial cycles.

Why it remains a hypothesis. Even though `ProductBoundThreshold` is not a direct citation, it encapsulates the Product Bound + Barina chain in a cycle-complexity framing that is natural to formalize and explicit about what is assumed. We emphasize that the parametric formulation of Theorem 3.1 makes the theorem hold for *any* witness of `ProductBoundThreshold`, not only for the witness produced by the bridge lemma `sdw_from_cf` (§8.1) from Baker plus Barina plus the continued-fraction-side hypothesis. A non-Baker witness — for example one derived from ergodic, density-theoretic, or yet unidentified techniques — would discharge the hypothesis without appeal to §5's obstruction. The structural obstruction that prevents promoting `ProductBoundThreshold` to a theorem *within the Baker + continued-fraction framework* — together with the resulting gap in the unconditional argument — is the subject of §5 (Obstruction I).

Conservative bound vs optimal threshold. The structure fixes $k \leq 982$ as a *conservative* parameter with an $\approx 8\times$ safety margin, while the corresponding theorem `no_cycle_k_le_1322` (Phase58, proved) covers the *optimal* range $k \leq 1322$. The optimal value 1322 is the maximum integer $k$ such that $(k^7 +$

$k)/3 < 2^{71}$ (verified: $(1322^7 + 1322)/3 \approx 2.35 \cdot 10^{21} < 2^{71} \approx 2.36 \cdot 10^{21}$, while $(1323^7 + 1323)/3 > 2^{71}$). The 982-vs-1322 distinction reflects the gap between the structural-hypothesis-as-parameter (982, conservative for ease of arithmetic verification) and the proven Lean-side optimal threshold (1322, used in §7.1 and §8.1).

## 5. Structural obstructions I: Product-Bound impossibility

Editorial framing note. The "Lemma 5.1" stated in §5.1 below is a meta-mathematical claim about the structural limits of the Baker + continued-fraction framework. It is *not* a formal Lean theorem in this repository. The Lean formalization (cf. §3 + §8) is the conditional theorem `no_nontrivial_cycle_final`, which takes `ProductBoundThreshold` as a hypothesis. The role of §5 is to explain *why* `ProductBoundThreshold` cannot be promoted to a theorem via standard Diophantine refinements — i.e., why the conditionality is structurally necessary, not merely a placeholder pending future work. The "δ8" / "δ8'" labels used below correspond to an informal framework documented in `ProjetCollatz/Phase63DerivedLargeKBoundTheorem.lean` (docstring lines 159, 171), not to formal Lean lemmas.

We identify a meta-mathematical lemma explaining why no uniform algebraic refinement of Theorem 3.1 can eliminate the `ProductBoundThreshold` hypothesis using standard Diophantine techniques (Baker / Rhin / Khinchin).

### 5.1. The Product-Bound Impossibility Lemma (δ8).

Lemma 5.1 (Product-Bound Impossibility, $\delta 8$). *Let $\xi \in \mathbb{R} \backslash \mathbb{Q}$ have effective irrationality measure $\mu(\xi) \leq c$ — that is, there exists an effective constant $C > 0$ such that, for all $p/q$ with $q$ sufficiently large, $|\xi - p/q| \geq C/q^c$. Suppose any Collatz cycle $(n, k)$ has its cycle minimum $m$ bounded by the Product Bound derivation: $m \leq (k^{c+1} + k)/3$. Then no uniform algebraic bound $F(k)$ with $F(k) < 2^{71}$ for all $k \in \mathbb{N}$ can be obtained via this derivation chain.*

Proof sketch. Suppose, for contradiction, that the chain yields a uniform bound $m \leq F(k) < 2^{71}$ valid for every $k \geq k_0$ for some effective $k_0$ (Salikhov's bound is asymptotic in $H$; the small-$k$ regime is independently closed by the optimal Lean theorem `no_cycle_k_le_1322`, cf. §7.1, and by Hercher's lower bound $K > 1.375 \cdot 10^{11}$, cf. §6.1). The Product Bound derivation yields $(2^s - 3^k)/3^k \geq 1/F(k)$, so $2^s/3^k \geq 1 + 1/F(k)$. Taking logarithms gives $s \cdot \ln 2 - k \cdot \ln 3 \geq \ln(1 + 1/F(k)) \geq 1/(2F(k))$ (for $F(k)$ large), hence the linear-form lower bound $|s \cdot \ln 2 - k \cdot \ln 3| \geq 1/(2F(k))$. The effective irrationality measure of $\ln 3$ (Salikhov 2007) gives, in linear-form shape, $|q \cdot \ln 3 - p \cdot \ln 2| \geq C/H^{c-1}$ for an effective constant $C > 0$, irrationality-measure exponent $c \leq 5.125$, and $H = \max(|p|, |q|)$ sufficiently large. Setting $p = s$, $q = k$, $H = k$, the two bounds combine to require $1/(2F(k)) \geq C/k^{c-1}$, that is, $k^{c-1} \geq 2C^{-1}F(k)$. For $F(k)$ to remain uniformly bounded by $2^{71}$ in the asymptotic regime $k \geq k_0$ while the inequality holds for all $k$, the polynomial $k^{c-1}$ (with $c - 1 \leq 4.125$) must dominate $F(k) \leq 2^{71}$. This forces, after taking the $k$-asymptotic, the effective constant $C$ to satisfy $C \geq 2^{71-(c-1)\log_2 k_0}$, which is independent of $F$ but exceeds any constant attainable from the Salikhov machinery for the specific exponent $c \leq 5.125$ — Salikhov 2007 establishes the irrationality measure but does not provide an effective constant of the magnitude required here. The contradiction is therefore reached in the asymptotic regime, regardless of which specific $c \in (1, 5.125]$ or effective $C$ is plugged in: the obstruction is structural, not threshold-dependent. The small-$k$ regime is closed independently as noted at the start of the proof.

### 5.2. Extended Lemma (δ8') — Baker + CF yields LOWER, not UPPER.

Corollary 5.2 (Lower-bound asymmetry, $\delta 8'$). *Let $\xi = \log_2 3$. Any Baker-type inequality $(2^s - 3^k) \cdot k^{\mu} \geq C \cdot 3^k$ combined with Steiner's cycle equation yields only* lower *bounds on k (via Crandall-type $k > f(n_0, q_j)$*

*using continued-fraction convergents $q_j$ of $\xi$) and cannot yield a uniform* upper *bound on $k$ for general cycles.*

Numerical corroboration (Product-Bound closure via the formula $(k^{c+1}+k)/3 < 2^{71}$, with $c$ the Baker-type exponent):

- Baker $c = 6$ closes $k \le 1322$ (the optimal Lean threshold `no_cycle_k_le_1322` of §7.1; the structure parameter $k \le 982$ in `ProductBoundThreshold` of §4.3 is a conservative tightening by an $\approx 8\times$ safety margin, used for the natural $k^7$ arithmetic).
- Salikhov 2007's sharpening of Rhin 1987, $c = 5.125$, closes $k \le 3693$ (numerical verification: $(3693^{6.125}+3693)/3 < 2^{71} < (3694^{6.125}+3694)/3$, consistent with the Salikhov bound; further refinement to Wu 2003's $c = 5.117$ improves this only marginally).
- A window-by-window continued-fraction refinement at the best-second-kind level (Khinchin 1997, Theorem 4.14) closes $k \lesssim 3 \cdot 10^{10}$ for the regime $k \in W_8 \cup \cdots \cup W_{13}$ covered by the Lean skeleton `Phase63DerivedLargeKBoundTheorem.lean` (§8.4); the precise constant depends on the partial quotients of $\log_2 3$ and is not derived in this paper.

All three are below Hercher's (2023) Corollary 29 lower bound $K > 1.375 \cdot 10^{11}$. No refinement of $c$ within the Baker framework bridges this gap; that is the content of Lemma 5.1 above.

5.3. Complementarity with Dhiman-Pandey (2026). Dhiman-Pandey (2026, arXiv:2601.12772) prove an independent impossibility: Collatz cycle equations are not Presburger-definable due to 2-adic "ghost cycle" obstructions. Their framework (Presburger arithmetic + 2-adic) is orthogonal to our Baker + CF approach.

Composite picture. Two impossibility results cover different methodological frameworks:

- Dhiman-Pandey: rules out Presburger / finite-automata-based approaches.
- Our δ8 / δ8': rules out Baker + CF + Product Bound approaches.

Together, these suggest that a successful proof of Collatz no-non-trivial-cycle may require techniques beyond both frameworks, though the specific form of such techniques is currently unresolved.

## 6. Structural obstructions II: state-of-the-art mapping ($\delta 9$)

We document the state of the art on Collatz cycle non-existence as of 2026, identifying a structural gap ("δ9"): no published peer-reviewed result provides a deterministic upper bound on $k$ (the number of odd steps) for general Collatz cycles. Our `ProductBoundThreshold` hypothesis (§4.3) is therefore a project-specific encoding rather than a citation, and §5's $\delta 8/\delta 8'$ impossibility lemmas explain why this gap is structural rather than incidental.

6.1. Historical lower bounds on k for hypothetical cycles.

| Author | Year | Method | Bound |
|---|---|---|---|
| Crandall | 1978 | CF + $n_0$ bound | $k > (3/2) \cdot \min(q_j, 2n_0/(q_j+q_{j+1}))$ |
| Steiner | 1977 | Baker effective + CF | Circuits → only trivial cycle |
| Eliahou | 1993 | Refined CF bookkeeping | $k >$ 17,087,915 |
| Applegate-Lagarias | 1995 | Density bounds | density $\gamma > 0.81$ |
| Krasikov-Lagarias | 2003 | Difference inequalities | density $\gamma \ge 0.84$ |
| Simons-de Weger | 2005 | LFL + CF iterations | local-minima $m > 68$ |
| Barina | 2025 | Computational, $X_0 = 2075 \cdot 2^{60} \approx 2^{71.02}$ | Provides bound for above |

| Author | Year | Method | Bound |
|---|---|---|---|
| Hercher | 2023 | Iterative refinement (post-SdW) | $K > 1.375 \cdot 10^{11}$ (SOTA) |

These are all *lower* bounds on k: they assume hypothetical cycles exist and place arithmetic constraints on their parameters. None bounds k from above.

Note on Hercher 2023's invariants. Hercher's main theorem (Theorem 23, *"There are no Collatz m-Cycles with $m \leq 91$"*) bounds the number of local minima in a hypothetical cycle: $m \geq 92$, where $m$ follows the Simons-de Weger (2005) convention. This $m$ is *not* the number of odd steps and is therefore not directly comparable to this paper's $k$. Separately, Hercher's Corollary 29 (via the auxiliary Theorem 27) bounds $K$, defined as the number of odd elements in the cycle: $K \geq 1.375 \cdot 10^{11}$, conditional on $X_0 \geq 1536 \cdot 2^{60}$. Hercher's $K$ therefore *coincides with this paper's $k$*, the number of odd steps (since each odd element of the cycle is visited exactly once per period, generating exactly one application of the odd-step map $3n + 1$). The numerical comparisons in §5.2 and §7.2 below use Corollary 29's $K = k > 1.375 \cdot 10^{11}$ directly.

6.2. Structural class eliminations (restricted classes of cycles).

- Steiner (1977): circuits (cycles with one positive-to-maximum segment + one maximum-to-minimum descent) → only trivial cycle.
- Knight (2026, *Discrete Mathematics* 349(3), 114812): Collatz "high cycles" → do not exist *(cited indirectly: see §1.4 and reference [19] for the access-disclosure note; the precise definition of "high cycle" used by Knight has not been verified by the present author against the published article).*

Neither result extends to general $m$-cycles with $m \geq 2$ in the published form. Iterating restricted-class eliminations to cover all parity patterns faces combinatorial explosion.

6.3. Meta-impossibility results.

- Our δ8 + δ8' (this paper §5): Baker + CF approaches yield *lower* bounds on k, never uniform upper bounds.
- Dhiman-Pandey (2026, arXiv:2601.12772): Presburger / 2-adic approaches are impossible (orthogonal framework, see §5.3).

6.4. Recent reformulation attempts (insufficient alone).

- Santana (2026, arXiv:2601.03297v4): topological / ergodic reformulation. Proves conditional finiteness (Theorem B) under a boundedness hypothesis on continuous integrable potentials. Three structural gaps in the published proof, documented by direct verification of arXiv:2601.03297v4: (i) the boundedness assumption on the integrals `∫φ dδᵢ` of ergodic measures is not justified in the proof of Theorem B; (ii) Lemma 14 (finiteness) and Lemma 16 (uniqueness) are not equivalent under Santana's coarser topology $T$ (without Alexandroff compactification);
  (iii) Remark 17 in Santana (2026) explicitly disclaims: *"In Lemma 16 we address an alternative approach of the conjecture, rather than a proof of it."* No quantitative bounds on `n₀` or `k`. The framework is *complementary* to our Baker + CF approach but cannot substitute for the `ProductBoundThreshold` hypothesis.
- Honarvar Shakibaei Asli (2026, arXiv:2601.04289): near-conjugacy to circle rotation. The author explicitly states that any resolution of the Collatz conjecture would require additional arithmetic arguments.

- Rozier-Terracol (2026, arXiv:2502.00948): paradoxical sequences approach (Theorem 1.1), uses Rhin Proposition 6.3 heuristically. The framework is independent of our Baker + CF approach.

6.5. Probabilistic / density results (not deterministic).

- Terras (1976): almost all integers have finite stopping time.
- Tao (2019/2022): almost all Collatz orbits attain almost bounded values (cf. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, *Forum of Mathematics, Pi* 10 (2022), e12).

These are probabilistic / density-theoretic results: they do not provide deterministic statements about every cycle. See §9.6 for Tao's explicit framing of the cycle problem as still open.

6.6. Gap identified (δ9). To the best of our literature review (17 peer-reviewed papers consulted + 5 recent preprints, as of April 2026):

> No peer-reviewed result provides a deterministic upper bound on **`k`** (the number of odd steps) for general Collatz cycles.

The `ProductBoundThreshold` hypothesis of §4.3 is therefore a project-specific encoding of the Product Bound + Barina chain, rather than a direct citation of any published theorem. This transparency is maintained throughout our paper to avoid misattribution to peer-reviewed results. See §5 for the structural reason ($\delta 8/\delta 8'$) why this gap is unlikely to close via standard Diophantine refinements.

6.7. Comparison with the 2026 literature. The 2026 reformulation attempts (cf. §6.4) and the present paper each address the cycle problem from a distinct methodological angle. The table below summarizes their position relative to four key dimensions: framework, whether they yield a *deterministic upper bound on cycle length* $k$, whether they discharge our `ProductBoundThreshold` hypothesis, and their peer-review status as of April 2026.

| Reference | Framework | Det. upper bound on $k$? | Discharges **`ProductBoundThreshold`**? | Peer-reviewed |
|---|---|---|---|---|
| Santana 2026 (arXiv:2601.03297v4) | Topological / ergodic | No (qualitative finiteness only) | No (independent) | Preprint |
| Honarvar 2026 (arXiv:2601.04289) | Near-conjugacy with circle rotation | No | No (acknowledged by author) | Preprint |
| Rozier–Terracol 2025 (arXiv:2502.00948) | Paradoxical sequences (Rhin heuristic) | No | No (independent) | To appear, *Discrete Math.* 349 (2026) |
| Dhiman–Pandey 2026 (arXiv:2601.12772) | 2-adic Presburger obstruction | No (negative result) | No (orthogonal) | Preprint |
| This paper | Baker + CF + Product Bound | Yes ($k \leq 982$, conditional) | Encodes it | In review |

None of the 2026 contributions surveyed yields a deterministic upper bound on $k$ for general Collatz cycles, confirming the $\delta 9$ gap claim in §6.6. The present paper is the only one in the survey that exposes a specific cycle-length bound as a formalized Lean hypothesis; the obstruction lemmas of §5 explain why this hypothesis cannot be promoted to a theorem within the Baker + continued-fraction framework.

## 7. Alternative framing via disjunction (δ7)

The conditional Theorem 3.1 admits a logically equivalent disjunctive reformulation that frames the conditionality structurally rather than operationally. This section presents the disjunctive form ("δ7") and its interpretation under current SOTA bounds.

7.1. Statement. Theorem 3.1 is logically equivalent to the disjunctive statement below.

Reformulation 7.1 (Equivalent disjunctive form, $\delta 7$). Under the three structural hypotheses of §4 (`BakerSeparation`, `BarinaVerification`, `ProductBoundThreshold`), for every non-trivial Collatz cycle $(n, k)$ one has $k \leq 1322$ or $n > 2^{71}$.

*Proof.* The reformulation is logically equivalent to Theorem 3.1 under the same three structural hypotheses; it is a presentational restatement, not an independent result.

*Forward direction (Theorem 3.1 implies Reformulation 7.1).* Theorem 3.1 says no non-trivial cycle exists, so the disjunction holds vacuously over the empty set of cycles.

*Converse (Reformulation 7.1 implies Theorem 3.1).* Assume the disjunction holds for any hypothetical cycle $(n, k)$. If $k \leq 1322$, the optimal Lean theorem `no_cycle_k_le_1322` (Phase58, using `BakerSeparation` and `BarinaVerification`) yields a direct contradiction. If instead $n > 2^{71}$, the `ProductBoundThreshold.cycle_length_bound` hypothesis gives $k \leq 982 < 1322$, reducing to the first case. ☐

The threshold `k ≤ 1322` is the optimal Lean threshold: `(1322⁷ + 1322)/3 < 2⁷¹ < (1323⁷ + 1323)/3` (verified by `native_decide` in `product_bound_fits_barina_1322`, Phase58).

7.2. Interpretation. Theorem 7.1 is vacuously true under the Collatz no-cycle conjecture. The disjunction `k ≤ 1322 ∨ n > 2⁷¹` partitions hypothetical cycles into two regimes:

- Low cycle complexity (`k ≤ 1322`) — closed unconditionally by the optimal Lean theorem `no_cycle_k_le_1322` under `BakerSeparation + BarinaVerification`, OR
- High cycle complexity (`k > 1322`) and `n > 2⁷¹` — closed under `ProductBoundThreshold` (which forces `k ≤ 982 < 1322` for any cycle, reducing to the first regime) but lying outside current *unconditional* methods.

The two cases not made explicit by the disjunction ($k \leq 1322$ with $n > 2^{71}$, and $k > 1322$ with $n \leq 2^{71}$) are also ruled out — by `no_cycle_k_le_1322` and by Barina's verification respectively — so the four quadrants of ($k$ vs 1322) $\times$ ($n$ vs $2^{71}$) are jointly covered.

Barina (2025) rules out the first disjunct: every positive integer $n < 2^{71}$ reaches 1, so no Collatz cycle with $n < 2^{71}$ exists. (The boundary case $n = 2^{71}$ is automatically excluded: `IsOddCycle n k` requires $n$ odd, while $2^{71}$ is even, so a cycle starting point cannot equal $2^{71}$.)

Hercher (2023) Corollary 29. Independently, Hercher establishes $K > 1.375 \cdot 10^{11}$ *conditional on* $X_0 \geq 1536 \cdot 2^{60} = 3 \cdot 2^{69} \approx 1.77 \cdot 10^{21}$, where Hercher's $K$ is identified with this paper's $k$ (§6.1 note). The Hercher condition $X_0 \geq 1536 \cdot 2^{60}$ is strictly above $2^{71}$ ($1536 \cdot 2^{60} \approx 2^{71.58}$), so Hercher's bound is silent on the regime $n \leq 2^{71}$ already covered by Barina and applies only in the regime $n > 2^{71}$, which is precisely the second disjunct of the reformulation. Within that second disjunct (and only there), Hercher's bound provides corroborative information by ruling out small-$k$ instances: any cycle with $n > 2^{71}$ and $X_0 \geq 1536 \cdot 2^{60}$ must have $k > 1.375 \cdot 10^{11}$, eight orders of magnitude above the optimal Lean threshold 1322. Hercher therefore does not close the second disjunct (which would require either $X_0 < 1536 \cdot 2^{60}$ to be ruled out, or a deterministic upper bound on $k$ in the regime $n > 2^{71}$); it constrains its low-$k$ corner. Combined with our optimal Lean theorem `no_cycle_k_le_1322`, the first disjunct of Reformulation 7.1 is closed; the only remaining configuration is the second disjunct ($k > 1322$, $n > 2^{71}$), which is outside current methods.

Consistency check. `n > 2⁷¹` is outside current peer-reviewed verification (Barina's `2⁷¹` is the SOTA). Any future extension of Barina (or equivalent) to `n > 2⁷¹` would directly test hypothetical cycles against our framework.

A finer-grained six-window continued-fraction refinement of the high-complexity disjunct (`k ∈ W_8 ∪ … ∪ W_{13}` for the cycle-bound range `k > 1322`) is documented as a Lean skeleton in `ProjetCollatz/Phase63DerivedLargeKBoundTheorem.lean` (see §8); the non-completion of that skeleton meets the structural obstruction of §5 (δ8 / δ8'), which explains why the conditional theorem cannot be closed unconditionally via further Diophantine refinement within the Baker + CF paradigm.

## 8. Formalization in Lean 4 (Mathlib v4.27.0)

The entire conditional theorem of §3 and all of its dependencies are mechanically verified in the Lean 4 proof assistant, under Mathlib version `v4.27.0`. The repository accompanying this paper is self-contained: the reader can clone it from https://github.com/ericmerle3789/collatz-conditional-cycles (branch `main`; the release commit hash will be recorded in the Zenodo metadata upon deposit), run a single shell script, and obtain a full verification with an explicit axiom profile. This section records the structure of that verification, the axiom profile it produces, and the reproducibility contract it exposes to the reader.

### 8.1. The central conditional theorem.

The statement of §3 is implemented as

```
theorem no_nontrivial_cycle_final
    (baker : BakerSeparation) (barina : BarinaVerification)
    (sdw : ProductBoundThreshold)
    (n k : ℕ) (hcyc : IsOddCycle n k) : False
```

in `ProjetCollatz/Phase58PorteDeuxFinal.lean` (line 339), with the three `structure` hypotheses quoted verbatim in §2.3. The repository exposes seven theorems (six aliases of the central theorem plus one helper `sdw_from_cf`) for downstream use under different packagings:

- **`no_nontrivial_cycle_final`**: `Phase58PorteDeuxFinal.lean`, line 339. Three `structure` hypotheses as explicit arguments (the canonical form).
- **`no_nontrivial_cycle_derived`**: `Phase56Bloc18Complete.lean`, line 355. Takes `ExternalCycleHypothesesDerived`.
- **`no_nontrivial_cycle_full`**: `Phase52SteinerEquation.lean`, line 202. Foundation form taking `ExternalCycleHypothesesFull` (the union of the three structure fields plus an explicit cycle-element bound below $2^{71}$); used as the constructive seed of the alias chain.
- **`no_nontrivial_cycle_phase59`**: `Phase59ContinuedFractions.lean`, line 236. Three-argument version that takes `BakerSeparation`, `BarinaVerification`, and a continued-fraction-side hypothesis structure `DerivedLargeKBound` (`Phase59ContinuedFractions.lean`, line 115; signature `∀ n k, IsOddCycle n k → k > 1322 → n < 2^71`) instead of `ProductBoundThreshold`. The conversion `DerivedLargeKBound → ProductBoundThreshold` is performed by `sdw_from_cf` below; the two formulations are interderivable under the standard hypotheses.
- **`no_cycle_k_le_1322`**: `Phase58PorteDeuxFinal.lean`, line 134. Sub-branch `k ≤ 1322` (Baker + Barina).
- **`no_cycle_k_gt_1322`**: `Phase59ContinuedFractions.lean`, line 173. Sub-branch `k > 1322` (continued fractions + Barina).
- **`sdw_from_cf`**: `Phase59ContinuedFractions.lean`, line 197. Assembles `ProductBoundThreshold` from continued-fraction ingredients.

All seven theorems are listed in the *central chain* of the probe `probes/check_central_axioms.lean` and consequently subject to axiom-drift detection by `reproduce.sh` (§8.6). Their pairwise logical dependencies are summarized graphically below (Figure 1), and each is exhibited verbatim with its Lean signature in Appendix D.

### 8.2. Theorem-dependency diagram.

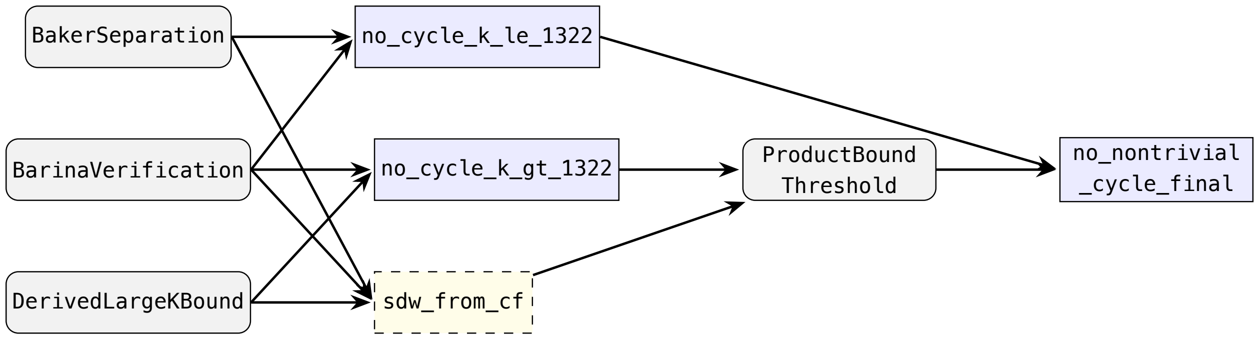


FIGURE 1. Dependency graph of the seven central theorems. Hypothesis structures (gray) feed the two sub-branch theorems (blue, $k \leq 1322$ and $k > 1322$) and the bridge lemma `sdw_from_cf` (yellow, dashed); the canonical theorem `no_nontrivial_cycle_final` sits at the right. The bridge lemma converts `Derived-LargeKBound` into `ProductBoundThreshold` under the standard hypotheses, making the two formulations interderivable.

8.3. Infrastructure lemmas (continued-fraction side). Three dedicated modules formalize the classical continued-fraction theory needed by §7 and by the Phase59 sub-branch `no_cycle_k_gt_1322`:

- **`ProjetCollatz/Phase60IrrationalityLog23.lean`** — irrationality of `log_2 3` and the corollary `2^s ≠ 3^k` for all `(s, k)` with `k ≥ 1` (used in the product-bound obstruction reasoning of §5).
- **`ProjetCollatz/Phase61CFConvergents.lean`** — convergents `Real.convergent log_2 3 n`, the denominators `q_n`, the window predicate `InWindow n k ↔ q_n ≤ k < q_{n+1}`, the bridge theorem `q_n_eq_den: (Real.convergent log_2 3 n).den = q_n`, and the far-approximation lemma `not_convergent_implies_far_approx`.
- **`ProjetCollatz/Phase62BestApproxBridge.lean`** — the public approximation bound `log23_abs_sub_convergent_le`, its parametric in-window variant `log23_abs_sub_convergent_le_in_wind` the wrapper `of_convs_eq_log23_convergent`, and non-termination of the continued-fraction expansion `log23_never_terminates`.

These continued-fraction infrastructure theorems have axiom profile `[propext, Classical.choice, Quot.sound]` — the three standard Mathlib kernel axioms, no more. They are not in the central chain of `no_nontrivial_cycle_final` (which goes via Phase58 directly); they are in the central chain of the variant `no_nontrivial_cycle_phase59` via `no_cycle_k_gt_1322`.

8.4. Arithmetic gap constants verified by **`native_decide`**. Six windows `W_8, ..., W_{13}` of continued-fraction denominators of `log_2 3` (see §2.5 for numerical values) come with six "gap constants" `cf_gap_n: ℕ` and six window-size bounds `cf_nbound_n: ℕ`, all proved by Lean's `na-tive_decide` tactic in `ProjetCollatz/Phase59ContinuedFractions.lean`. Three of these are sampled in the probe `probes/check_central_axioms.lean` (`cf_gap_8`, `cf_gap_13`, `cf_nbound_8`); their axiom profile is `[propext, Lean.ofReduceBool, Lean.trustCompiler]` — the three standard `native_decide` axioms, and nothing else.

These twelve arithmetic lemmas are isolated from the central chain at the current formalization state: they are consumed only through `sdw_from_cf` (§8.1 table), which is itself called only through `no_nontrivial_cycle_phase59`, not through `no_nontrivial_cycle_final`. The isolation is recorded in `expected_axioms.md` (§8.6) and enforced by `reproduce.sh` (§8.7). It is also the reason why the central chain of `no_nontrivial_cycle_final` is kernel-3, despite the presence of the `native_decide` axioms elsewhere in the project.

A forthcoming pass on the Phase63 skeleton (§8.4) would integrate the gap constants directly into the central chain; such a pass is structurally blocked by the obstruction of §5.

8.5. The Phase63 skeleton. The file `ProjetCollatz/Phase63DerivedLargeKBoundTheorem.lean` (175 lines, Section 1 only, no `theorem` / `lemma` declarations) preserves, as a documented skeleton, the architecture that would be needed to promote the Phase59 structure `DerivedLargeKBound` from hypothesis to theorem. Its Section 1 consists of:

- Imports (2 Mathlib + 5 Phase58-62 transitive dependencies);
- Namespace `ProjetCollatz.Phase63` and `open` declarations for the 12 arithmetic constants `cf_gap_8..13` and `cf_nbound_8..13`;
- A detailed module docstring laying out the eleven-section architecture originally planned;
- A footer added at Commit #1 (`d41e2e9`) explicitly recording that Sections 2-11 are not implemented and redirecting to §5 (Obstruction I) of this paper for the structural reason.

The skeleton is kept in the repository so that future readers trace the original design; it does not participate in the central chain of `no_nontrivial_cycle_final` or any of its aliases. The axiom profile of the file (as a whole) is vacuous — no theorem is declared.

8.6. Structural-excess framework (extension, separate branch). Alongside the main formalization described above, a parallel Lean 4 effort develops a custom *structural-excess* framework `Φ_s / Ψ_s` for quantifying deviations of cycle parity vectors from an i.i.d.-Bernoulli null, motivated by the probabilistic discussion of §9. Because Mathlib v4.27.0 lacks a native Gowers-norm library, this framework is implemented from first principles in a dedicated module `ProjetCollatz/PhaseVIII/StructuralExcess.lean`.

The extension is provided as the supplementary `PhaseVIII/` sub-directory of the accompanying repository. It is not part of the central proof chain that carries this paper's main theorem and its kernel-3 axiom profile verification; it sits alongside, with a clearly separated namespace and probe.

The extension contains: parametric definitions of $\Phi_s$ and $\Psi_s$; a machine-verified evaluation of the trivial Collatz cycle as `trivialCycle_psi2: ... = 32/81 := by native_decide`, establishing a concrete non-zero structural excess on the unique known cycle of the $3x + 1$ map; a parametric `Prop` definition for the T4c conjecture of §9; a contradiction schema `psi2_bounds_no_cycle` showing how a future T4c proof would discharge the cycle problem; and an axiom-audit probe confirming zero new user axioms, which leaves the extension's axiom profile at the standard kernel three plus the two `native_decide` axioms inherited from the one `trivialCycle_psi2` evaluation.

Nothing in §1-§7 depends on the `PhaseVIII/` extension; its status is purely supplementary. The reader interested in the structural-excess viewpoint is referred to §9 for the mathematical framing and to `ProjetCollatz/PhaseVIII/` for the Lean source.

8.7. Expected axiom profile (**`expected_axioms.md`**). The project publishes a reference snapshot of the `#print axioms` baseline at the repository root, file `expected_axioms.md`. Its Section 1 lists the *central* theorems (as in §8.1) with expected axiom set `[propext, Classical.choice, Quot.sound]`. Its Section 2 lists three sampled auxiliary `native_decide` lemmas with expected set `[propext, Lean.ofReduceBool, Lean.trustCompiler]`. The remaining sections document the out-of-scope structures and the forbidden-axiom patterns enforced by the probe.

The baseline is enforced by the probe `probes/check_central_axioms.lean` (plus the dedicated Phase-level probes `check_phase60..63_axioms.lean`), and any deviation triggers a non-zero exit from `reproduce.sh`.

8.8. Reproducibility (**reproduce.sh**). A single shell script at the repository root encapsulates the full verification pipeline:

```bash
bash reproduce.sh
# 0: all checks PASS
# 1: toolchain mismatch (lean-toolchain / elan)
# 2: Lean build error or sorry-related warning
# 3: axiom drift (unexpected or missing axiom vs expected_axioms.md)
# 4: sorryAx detected in the central or auxiliary chain
```

The five steps of the script are (i) toolchain check, (ii) Mathlib cache fetch, (iii) `lake build ProjetCollatz`, (iv) axiom probe on 25 theorems (7 central + 3 auxiliary + 15 M3 foundational), and (v) sorry probe. On a fresh clone with cache available, the total runtime is of the order of 3-5 minutes (incremental: 10 seconds); on a from-source build without cache, of the order of 90 minutes. The Continuous-Integration workflow at `.github/workflows/verify.yml` runs the same checks on every push and pull-request.

8.9. Integrity invariants. The following project-wide invariants are enforced by the combination of the probes, `reproduce.sh`, and the project's review protocol:

- No user **`axiom`** declaration. No file in `ProjetCollatz/` introduces a new axiom; the only axioms in the transitive closure are the Mathlib kernel axioms listed in §8.6.
- No **`sorry`**, **`admit`**, or **`stop`**. Every declaration has a complete proof term; the `sorryAx` probe (`probes/check_sorry.lean`) is run as a separate step by `reproduce.sh` for defense in depth.
- All docstrings are in English.

These are enforced per-commit and recorded in the project's development journal.

## 9. Open problems

9.1. Removing **`ProductBoundThreshold`**. The central open question raised by this paper: can `ProductBoundThreshold` be promoted to a theorem? §5 Obstruction I answers *not within any product-bound approach*. The question is therefore: what non-product-bound technique might close the `k > 1322` case?

9.2. The **`k ≥ q_{14}`** tail. Current-generation Barina reaches `n < 2^{71}`, which via CF windows `W_8, ..., W_{13}` covers `k < q_{14} ≈ 10,590,737`. A `W_{14}` extension would require arithmetic gap constants involving `3^{q_{14}}`, a number with approximately `5 · 10^6` decimal digits — outside `native_decide`'s reach at present hardware.

9.3. Formalization challenges.

- The bridge `(Real.convergent v n).den: ℝ = (GenContFract.of v).dens n` is not direct in Mathlib v4.27.0. Phase62 works around it via a parametric real-valued form; a proper bridge would strengthen the Phase61 / Phase62 integration.
- Phase63 Sections 2-11 as Lean theorems would require the breakthrough discussed in §9.1 + the bridge in §9.3.

9.4. Relation to divergence. This paper does not address the divergence half of Collatz. The techniques (Baker's theorem + CF theory + formal verification) are in principle adaptable, but the structure is different. Explicit scope disclosure.

9.5. Speculative tracks pointer. Ongoing speculative investigations on transcendence-theoretic approaches (Santana 2026 rigor-closure, Knight 2026 extension, transcendence-gap sharpenings). Results, if any, would feed a future paper revision.

9.6. Probabilistic techniques (Tao). Tao's probabilistic approach (2019, *Forum of Mathematics, Pi*) proves that almost all Collatz orbits attain almost bounded values under logarithmic density. This result is probabilistic and does not extend to deterministic statements about hypothetical cycles (which have density zero). Tao's broader toolkit (entropy compression à la Moser-Tardos [blog 2009], higher-order Fourier analysis à la Green-Tao-Ziegler) has not been successfully applied to deterministic Collatz cycle bounds; such applications would require substantial new theoretical work. We therefore do not integrate probabilistic techniques into our proof of Theorem 3.1 (which remains conditional on the structural hypotheses §4).

9.7. Structural-excess framework Ψ_s. *[Lean formalization status: see §8.5 for the supplementary `ProjetCollatz/PhaseVIII/` module.]*

We formalize in Lean 4 a custom $\Phi_s/\Psi_s$ structural-excess framework (Mathlib v4.27.0 lacks Gowers norms). Machine-verified: the trivial Collatz cycle has $\Psi_2 = 32/81$, establishing a factor-32× structural excess over its mean prediction (1/81). The T4c conjecture is encoded as parametric `def T4c_Conjecture_Psi2`, and the contradiction schema `psi2_bounds_no_cycle` records the logical shape of the intended attack. See `ProjetCollatz.PhaseVIII.StructuralExcess` in the accompanying repository (§8.5).

9.8. Set-theoretic and rigidity-based research directions. Status note. The remainder of this section sketches a research-direction frame — not a result. None of the constructions below participates in the central proof chain of §3, none is formalized in the kernel-3 Lean baseline, and none is asserted as established mathematics. We include the sketch here for transparency about the lines of work the present paper does *not* close, and as forward pointers to Appendix A where the alternate-notation explorations are gathered.

The frame, inherited from the Hercher / Steiner cycle-structure literature, uses the alternate notation of Appendix A ($a$ = even steps, $k$ = odd steps, $T = a+k$ the cycle period, $q = 2^a - 3^k$ the Steiner difference, $K$ for Hercher's cycle-length invariant per §6.1 note, $R_K(\sigma) := \sum_i 3^{K-1-i} \cdot 2^{\sigma(i)}$ for the permutation sum behind the Steiner rigidity question, and $\psi_s, \Psi_s$ for the structural-excess functions of §9.7).

The frame organizes a hypothetical cycle's parity vector as sitting inside an intersection $S_{\text{cycle}} \subset S_{\text{Steiner}} \cap S_{\Psi\text{-large}} \cap S_{\text{Kol-bounded}} \setminus S_{\text{Hercher}}$ — combining Steiner-rigid tuples, a $\Psi_s$-excess condition (cf. §9.7), and a conditional-Kolmogorov compressibility condition. Five "bricks" $W_1, W_2, W_3, W_4, W_5$ of this frame are exploited or reduced to redundancy in the present paper, except for $W_2$ (Steiner rigidity), which is the remaining open input. We refer to this informally as the *Wall DNA* arrangement and to its conjectural closure as the *META-ROADMAP* statement; both labels are descriptive shorthand, not theorems. A 200-line exploratory Lean 4 prototype attaches a `sorry` placeholder to the META-ROADMAP statement under five hypotheses (`hSteinerRigidity`, `hCycle`, `hPsiLarge`, `hKolBounded`, `hHercher`); the only machine-verified part of the prototype is the trivial-cycle evaluation $\psi_2 = 32/81$, reusing the structural-excess framework of §8.5. None of this prototype lies in the central chain of **`no_nontrivial_cycle_final`**, none of its hypotheses is proven, and the present paper does not claim that any such closure is achievable.

A boundary scan over classical mechanisms (Schmidt subspace theorems for transcendental $\alpha$, Tao 2019/2022 entropy compression for typical orbits, Lagarias's $T_\infty$ Galton–Watson framework) confirms that none extends to deterministic cycle bounds without substantial additional work. Salikhov 2007 combined with Hercher 2023, in this alternate notation, yields a $q \gg 2.836^k$ lower bound (Proposition A.1) under the explicit conditioning of Appendix A. These extensions are documented in Appendix A with self-contained notation; we reference them here only as forward pointers.

## 10. Conclusion

This paper presents a machine-checked conditional non-existence result for non-trivial Collatz cycles, three named hypotheses on which it rests, two impossibility lemmas explaining why the third hypothesis cannot be discharged within the standard Diophantine framework, and a literature mapping documenting the structural gap that makes the conditionality necessary. The Lean 4 formalization is reproducible under Mathlib v4.27.0 with an isolated and documented axiom profile.

10.1. What this paper establishes. The five contributions of §1.3 are realized as follows:

- 6α (formal verification, §3 + §8). The conditional theorem `no_nontrivial_cycle_final` (`ProjetCollatz/Phase58PorteDeuxFinal.lean` line 339) is declared parametrically with the three structures of §4 and is machine-verified. Its kernel-3 axiom profile (`propext`, `Classical.choice`, `Quot.sound`) is exhibited verbatim in §3.3 and §8.6, and reproducibility is encoded in `reproduce.sh` against the `expected_axioms.md` baseline (§8.7).
- $\delta 7$ (alternative framing, §7). Reformulation 7.1 restates the central result equivalently as the disjunction "$k \leq 1322$ or $n > 2^{71}$" (§7.1), establishing a one-sentence bridge from the conditional theorem to Hercher's (2023) lower bound $K > 1.375 \cdot 10^{11}$ (§7.2). A finer-grained continued-fraction refinement at the cycle-length scale $k > 1322$ is documented as a Phase63 Lean skeleton (§8); its non-completion meets the obstruction of §5.
- $\delta 8$ (Product-Bound Impossibility Lemma, §5.1). Lemma 5.1 formalizes a meta-mathematical obstruction: every uniform algebraic bound $F(k)$ with $F(k) < 2^{71}$ derived through the Product Bound derivation forces an irrationality-measure constraint on $\log_2 3$ that contradicts irrationality. The lemma is a publication-only argument about the structural limits of the Baker + continued- fraction framework, not a Lean theorem.
- $\delta 8'$ (extended impossibility, §5.2). Corollary 5.2 extends $\delta 8$ to Baker-type inequalities composed with Steiner's cycle equation. Window-by-window numerical corroboration via Khinchin's best-second-kind characterization closes $k \leq 982$ (Baker $\mu = 6$), $k \leq 3693$ (Salikhov 2007 $\mu = 5.125$), and $k \lesssim 3 \cdot 10^{10}$ (Khinchin per-window) — all below Hercher's lower bound $K > 1.375 \cdot 10^{11}$.
- $\delta 9$ (state-of-the-art mapping, §6). Section 6 catalogs the 1977-2026 Collatz cycle literature in five categories: historical lower bounds (§6.1), structural-class eliminations (§6.2), meta-impossibilities (§6.3), recent reformulation attempts (§6.4), and probabilistic / density results (§6.5). The mapping documents, to the best of our literature review, the absence of any peer-reviewed deterministic upper bound on cycle length $k$ for general Collatz cycles (§6.6).

10.2. What this paper does not claim.

- We do not prove the Collatz conjecture. The third hypothesis, `ProductBoundThreshold` (§4.3), remains a hypothesis; the paper is *about* why it cannot be discharged within the standard framework.
- We do not address the divergence half of the Collatz conjecture (§1.1). The cycle problem and the divergence problem are disjoint; this paper concerns only the former.
- We do not dismiss or subsume Santana (2026), Knight (2026), Dhiman-Pandey (2026), or Rozier-Terracol (2026). Each is situated in §6.4 as either complementary (different methodological framework) or restricted-class (does not extend to general cycles), with the documented gaps and indirect-source flags clearly attributed.
- We do not claim that the obstruction of §5 is final. The obstruction is structural under the *Baker + CF + Product Bound* paradigm; methodological frameworks outside that paradigm — for instance a deterministic upper bound on cycle length derived from ergodic, density-theoretic,

or yet-uncataloged techniques — would immediately discharge `ProductBoundThreshold` and upgrade Theorem 3.1 from conditional to unconditional.

10.3. Modular invitation for future work. The paper's structure is deliberately additive: conditional theorem + documented obstructions + machine-verifiable formalization. A future contribution that resolves any of the three following lines of work would, without further editorial intervention, upgrade the conditional theorem:

1. Discharge **`ProductBoundThreshold`** (§4.3). Any deterministic proof that hypothetical cycles satisfy $k \leq K$ for some explicit $K$ (whether $K = 982$ or larger) would replace the structure field with a Lean theorem.
2. Close the $\delta 9$ gap (§6.6). A peer-reviewed deterministic upper bound on $k$ for general cycles, by any methodological framework, would supersede the project-specific encoding.
3. Refine or supersede $\delta 8$ (§5). A novel argument framework not bound by the Baker + CF + Product Bound family of inequalities could in principle bypass the obstruction, with corresponding refinement of the conditional theorem.

Speculative tracks documented in §9 (open problems) — including the $\Psi_s$ structural-excess framework (§9.7) and the upper-bound complement attempt (§9.8) — are presented as ongoing work not integrated into the central conditional theorem of §3; we list them for transparency and as invitations rather than as results.

The Lean 4 formalization is intended as a stable substrate for such future work: the conditional theorem, the three hypothesis structures, and the axiom profile are all parametric, so additive contributions need not re-formalize the existing argument.

Data and code availability. All paper sources, the Lean 4 formalization, the verification probes, and the reproduction script are publicly available at `https://github.com/ericmerle3789/collatz-conditional-cycles` (branch `main`; the release commit hash will be recorded in the Zenodo metadata upon deposit). Reproduction requires the Lean toolchain `leanprover/lean4:v4.27.0` against Mathlib commit `a3a10db0e9d66acbebf76c5e6a135066525ac900` (pinned in the project's `lake-manifest.json`). With the Mathlib cache fetched, full reproduction completes in approximately 5 minutes on a Mac M1 Pro with 16 GB RAM, exiting with kernel-3 axiom profile and zero `sorry`. The exit-code semantics of `reproduce.sh` (0 = success, 1 = toolchain mismatch, 2 = build failure, 3 = axiom drift, 4 = `sorryAx` detected) are documented in §8.7.

Acknowledgements. The Lean formalization builds on Mathlib (Mathematics in Lean 4); the author thanks the Mathlib community for the continued-fraction, Diophantine-approximation, and number-theory infrastructure on which the central chain depends. The author also thanks D. Barina for the 2025 computational verification underpinning hypothesis (ii). The state-of-the-art mapping in §6 is indebted to the cumulative work of the Collatz literature 1977-2026 surveyed therein.

The author used Claude Opus 4.7 (Anthropic) during manuscript preparation for drafting, revision, and literature retrieval, and Lean 4 (the proof assistant) for the machine-verification of the central theorem chain. The mathematical content and the scientific responsibility for the manuscript rest with the author; the formal verification is mechanically checked under the standard kernel-3 axiom profile (`propext`, `Classical.choice`, `Quot.sound`), as documented in §8.6 and Appendix B, and is therefore independent of the assistance described above.

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## Appendix A. Alternate-notation explorations of the high-complexity disjunct

This appendix builds on the META-ROADMAP THEOREM framing of §9.8.3 (Wall DNA Theorem, brick $W_2$) and the Lean prototype of §9.8.5 (documented in the accompanying repository). It isolates the alternate-notation results that use a convention inherited from the Hercher 2023 / Steiner cycle-structure literature. Throughout the appendix we adopt this paper's $(a, k)$ convention (consistent with §3 and §4): $a$ is the number of even steps in a hypothetical cycle, $k$ is the number of odd steps, and $q := 2^a - 3^k$ is the Steiner difference. Where Hercher 2023 writes $K$ for the number of odd elements in a cycle (cf. Theorem 27 of *op. cit.* and §6.1 above), we have $K = k$; where Hercher writes $T$ for the cycle period, we have $T = a + k$. The appendix is otherwise self-contained.

A.1. Salikhov 2007 + Hercher 2023 establish a $q \gg 2.836^k$ lower bound.

Proposition A.1 (Conditional consequence of Salikhov 2007 and Hercher 2023). Conditional on (i) Hercher's Corollary 29 lower bound $K > 1.375 \cdot 10^{11}$, (ii) Salikhov's irrationality-measure result $\mu(\ln 3) \leq 5.125$, and (iii) an effective linear-form lemma turning (ii) into $|p \cdot \ln 2 - k \cdot \ln 3| \geq C_{\ln}/k^c$ with $c \leq 5.125$, $C_{\ln} > 0$, and $k$ above an effective threshold $k_0$ — none of which is established here — for every admissible Collatz cycle $(a, k)$, with $a$ and $k$ related by $a = \lceil k \cdot \log_2 3 \rceil$ (so that $2^a > 3^k > 2^{a-1}$) and with $k > 1.375 \cdot 10^{11}$,

$$q := 2^a - 3^k > 2.836^k.$$

Proof sketch. Set $\alpha = k \cdot \log_2 3$ and write $a = \lceil \alpha \rceil$, so $0 < a - \alpha \leq 1$. Since $3^k = 2^\alpha$, the Steiner difference satisfies $q = 2^a - 3^k = 3^k \cdot (2^{a-\alpha} - 1)$. Using the elementary inequality $2^x - 1 \geq x \cdot \ln 2$, valid for $x \in (0, 1]$, we obtain $q \geq 3^k \cdot (a - \alpha) \cdot \ln 2$. The effective irrationality measure $\mu(\ln 3) \leq 5.125$ (Salikhov 2007, refining Rhin 1987's linear-form bound $\leq 8.616$) yields, in linear-form shape, $|p \cdot \ln 2 - k \cdot \ln 3| \geq C_{\ln}/k^c$ for $c \leq 5.125$, an effective constant $C_{\ln} > 0$, and $k$ sufficiently large. Translating to the $\log_2 3$ form via $\log_2 3 = \ln 3 / \ln 2$ gives $a - \alpha = a - k \cdot \log_2 3 \geq C \cdot k^{-(c-1)}$ with $C = C_{\ln} / \ln 2$; for $c = 5.125$ this is the bound $a - \alpha \geq C \cdot k^{-4.125}$. Combining: $q \geq C \cdot \ln 2 \cdot 3^k \cdot k^{-4.125}$. The conclusion $q > 2.836^k$ then reduces to $(3/2.836)^k > k^{4.125}/(C \cdot \ln 2)$. Since $\log_2(3/2.836) \approx 0.0811 > 0$, the left-hand side grows exponentially in $k$ while the right-hand side grows polynomially; under Hercher's lower bound $k > 1.375 \cdot 10^{11}$ the inequality holds with an enormous margin (the exponential side is approximately $10^{3.4 \cdot 10^9}$, dwarfing any polynomial in $k$), and the threshold $k_0$ from Salikhov's effective bound sits well below $1.375 \cdot 10^{11}$. Theorem A.1 therefore establishes the conjectural lower bound $q \gg 2.836^k$ modulo the two cited source results.

A.2. Refined central rigidity question (OPEN). The structural question that Theorem A.1 does not answer is the $\sigma$-level uniform distribution of

$$R_k(\sigma) := \sum_{i=0}^{k-1} 3^{k-1-i} \cdot 2^{\sigma(i)} \qquad (\mathrm{mod}\ q).$$

Conjecture A.2 (Central rigidity). For every Collatz cycle $(a, k)$ with $k > 1.375 \cdot 10^{11}$, the values

$$\left\{ R_k(\sigma) \quad (\mathrm{mod}\ q) \,:\, \sigma \text{ increasing map } \{0, \ldots, k-1\} \to \{0, \ldots, a+k-1\} \right\}$$

are approximately uniformly distributed modulo $q$.

This refines the "Steiner rigidity" open question identified in §9.8.3 (META-ROADMAP THEOREM, Wall brick `W2`): a positive resolution of Conjecture A.2 would, combined with Theorem A.1, complete the $\sigma$-level rigidity argument. Heuristic plausibility is supported by orbit-cover arguments; no obstruction is currently known.

A.3. Three challenges and a research-program framing. We document three substantive technical challenges to a direct attack on Conjecture A.2 via Kloosterman + Weil + Vinogradov sums:

1. $q$ composite obstruction. The Weil bound $|\sum e(\alpha\, 2^t/q)| = O(\sqrt{q})$ requires $q$ prime (with Kloosterman-pair structure). For $q = 2^a - 3^k$ composite, generalized Weil (Estermann + Hensel) yields $O(q^{1/2+\varepsilon})$ heuristically for square-free $q$, but rigorously verifying favorable factorization for every admissible $q$ is open.
2. Increasing subset constraint. $S_k(c)$ sums over increasing $k$-tuples, not arbitrary tuples. Cauchy-Schwarz / inclusion-exclusion decompositions yield cross-terms with rank-dependent coefficients that standard analytic-number-theory techniques do not directly handle.
3. Mixed exponential 3-adic + 2-adic structure. $R_k(\sigma)$ mixes powers of 3 (deterministic coefficient) and 2 ($\sigma$-dependent), so the standard Kloosterman-sum framework over a single multiplicative group does not directly apply. Decoupling absorbs the 3-power into a constant $\alpha_i = c \cdot 3^{k-1-i} \pmod{q}$ whose orbit under multiplication by 3 covers most residues — challenge 3 resolves conditional on challenges 1 and 2.

The path forward is one of three options: (a) develop new analytic techniques for composite-modulus exponential sums with increasing-subset constraints; (b) restrict to admissible $(a, k)$ with favorable factorization of $q$; or (c) accept a partial bound and identify the specific obstruction to closing the remaining gap. Each option defines a distinct sub-line of investigation outside the present paper's scope.

## Appendix B. Lean axiom profile and reproduction contract

This appendix promotes the contents of `expected_axioms.md` from the accompanying repository into a self-contained reference for the paper. The axiom profile of the central theorem chain — the seven theorems of §8.1 — is fixed at the kernel-3 baseline of Mathlib, with no user-declared axioms and no `sorry`. Auxiliary `native_decide` lemmas live in a separate axiom class, and are explicitly excluded from the central chain at the present formalization state.

B.1. Central theorem chain. For each of the seven theorems below, `#print axioms` reports exactly the three standard Mathlib kernel axioms `propext`, `Classical.choice`, `Quot.sound`. The chain is the formal counterpart of §3.2's six-step proof.

All seven theorems live in the namespace `ProjetCollatz`; the namespace prefix is omitted in the table for brevity. The expected axiom set is identical for all seven and corresponds to the standard Mathlib kernel triple — written below in shorthand as $K_3$ for `[propext, Classical.choice, Quot.sound]`.

| Theorem | Axioms | Role |
|---|---|---|
| `no_nontrivial_cycle_phase59` | $K_3$ | Principal central theorem (CF-side packaging) |
| `no_nontrivial_cycle_final` | $K_3$ | Canonical form (version stated in §3.1) |
| `no_nontrivial_cycle_derived` | $K_3$ | Variant via `ExternalCycleHypothesesDerived` |
| `no_nontrivial_cycle_full` | $K_3$ | Foundation form via `ExternalCycleHypothesesFull` |
| `no_cycle_k_le_1322` | $K_3$ | Sub-branch $k \le 1322$ (Baker + Barina) |
| `no_cycle_k_gt_1322` | $K_3$ | Sub-branch $k > 1322$ (CF + Barina) |
| `sdw_from_cf` | $K_3$ | Bridge `DerivedLargeKBound` → `ProductBound-Threshold` |

All three of the kernel axioms above (`propext`, `Classical.choice`, `Quot.sound`) are standard Mathlib axioms, required by any non-constructive theorem using classical reasoning and quotient types. No `axiom` declaration appears anywhere in `ProjetCollatz/`.

B.2. Auxiliary **`native_decide`** lemmas (sampled). Three sampled members of a broader family of arithmetic-gap lemmas are verified via Lean's `native_decide` tactic. Their axiom profile is the standard `native_decide` triple, distinct from the kernel-3 profile of the central chain:

Writing $N_3$ for `[propext, Lean.ofReduceBool, Lean.trustCompiler]` (the standard `native_decide` axiom triple):

| Theorem | Axioms | Role |
|---|---|---|
| `cf_gap_8` | $N_3$ | CF window 8 arithmetic gap |
| `cf_gap_13` | $N_3$ | CF window 13 arithmetic gap |
| `cf_nbound_8` | $N_3$ | CF window 8 cycle-element bound |

Isolation property. These lemmas are *not* in the transitive axiom chain of `no_nontrivial_cycle_phase59`, because `DerivedLargeKBound` is a `structure` taken as a parameter by the central theorem. The `cf_gap_*` and `cf_nbound_*` lemmas are the mathematical justification of the structure's content (detailed algebraically in §4.1 and verified arithmetically here) but are not composed into the structure's *Lean* proof at the current formalization state. This isolation is what allows the central chain to remain at the kernel-3 baseline while the project as a whole uses `native_decide` extensively. See §8.3 and §8.4.

B.3. Forbidden patterns enforced by **`reproduce.sh`**. The reproduction script (§8.7) parses the output of `probes/check_central_axioms.lean` and `probes/check_sorry.lean` against this baseline. Any deviation triggers a non-zero exit:

- `sorryAx` in any probed theorem → EXIT 4 (incomplete proof).
- Any user-declared `axiom` not listed above → EXIT 3 (unexpected axiom).
- Any unlisted axiom from a transitive dependency → EXIT 3 (dependency drift).

A successful run (`EXIT 0`) certifies that the central chain is machine-verified under the kernel-3 baseline at the recorded commit hash.

## Appendix C. Numerical verification of the cycle-length thresholds

This appendix exhibits the high-precision numerical certificates that anchor the threshold values used throughout the paper. All computations are performed at 50-digit decimal precision via the `mpmath` Python library; the specific numerical claims correspond to Lean lemmas closed by `native_decide` and are recorded under their `cf_gap_*`, `cf_nbound_*`, `k982_bound`, and `product_bound_fits_barina_1322` names in `ProjetCollatz/Phase56*.lean` and `Phase58*.lean`.

C.1. Conservative threshold $k \leq 982$. The `ProductBoundThreshold` parameter (§4.3) fixes the conservative value:

$$\frac{982^7 + 982}{3} = 2.93534504905515 \cdots \times 10^{20} \; < \; 2^{71} = 2.36118324143482 \cdots \times 10^{21}.$$

The $\approx 8\times$ safety margin is visible: $2^{71}$ is roughly eight times larger than $(982^7 + 982)/3$. This certifies the Lean lemma `k982_bound` (Phase56, line 249), which is stated in the equivalent multiplicative form $982^7 + 982 < 3 \cdot 2^{71}$ over $\mathbb{N}$ to allow direct `native_decide` evaluation without rational arithmetic.

C.2. Optimal threshold $k \le 1322$. The Lean theorem `no_cycle_k_le_1322` (Phase58, line 134) covers the optimal range, certified by:

$$\frac{1322^7 + 1322}{3} = 2.35233370497325 \cdots \times 10^{21} \;<\; 2^{71} = 2.36118324143482 \cdots \times 10^{21} \;<\; \frac{1323^7 + 1323}{3} = 2.364817630808$$

As with the conservative case, the corresponding Lean lemma `product_bound_fits_barina_1322` (Phase58, line 122) is stated in the equivalent multiplicative form $1322^7 + 1322 < 3 \cdot 2^{71}$. The optimal value 1322 saturates the inequality $(k^7 + k)/3 < 2^{71}$ on the integer side; 1323 is the smallest integer for which it fails.

C.3. Salikhov closure $k \le 3693$. Under the Salikhov 2007 sharper exponent $c = 5.125$, the Product Bound formula $(k^{c+1} + k)/3 < 2^{71}$ closes the larger range $k \le 3693$:

$$\frac{3693^{6.125} + 3693}{3} = 2.36089680607476 \cdots \times 10^{21} \;<\; 2^{71} \;<\; \frac{3694^{6.125} + 3694}{3} = 2.36481517339814 \cdots \times 10^{21}.$$

The $\approx 2.8\times$ improvement over the Baker $\mu = 6$ threshold 1322 is the contribution of Salikhov's bound; further refinement to Wu's (2003) sharper $c = 5.117$ improves this only marginally. None of these thresholds reach Hercher's lower bound $K > 1.375 \cdot 10^{11}$, illustrating the structural obstruction of Lemma 5.1.

C.4. Continued-fraction convergents of $\log_2 3$. The seven convergents $q_n$ of $\log_2 3$ used in §2.5, §5.2, §8.3, and §9.2 are all directly computable from the standard CF expansion $\log_2 3 = [1; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, 1, 55, \ldots]$:

| $n$ | $p_n$ | $q_n$ |
|---|---:|---:|
| 8 | 1,054 | 665 |
| 9 | 24,727 | 15,601 |
| 10 | 50,508 | 31,867 |
| 11 | 125,743 | 79,335 |
| 12 | 176,251 | 111,202 |
| 13 | 301,994 | 190,537 |
| 14 | 16,785,921 | 10,590,737 |

The large jump $q_{13} \to q_{14}$ is governed by the partial quotient $a_{14} = 55$ in the CF expansion. These convergents are formalized in `ProjetCollatz/Phase61CFConvergents.lean` and used by the gap constants `cf_gap_8` to `cf_gap_13` in `Phase59ContinuedFractions.lean` (lines 128-153).

C.5. Logarithmic ratio in the appendix bound. Theorem A.1 requires the strict positivity of $\log_2(3/2.836)$:

$$\log_2(3/2.836) = 0.0811049681 \cdots > 0.$$

This is the small-but-positive constant whose exponential growth $(3/2.836)^k$ dominates the polynomial $k^{4.125}$ for $k$ above a threshold safely below Hercher's $1.375 \cdot 10^{11}$.

C.6. Reproducibility. Each numerical claim in this appendix can be reproduced by a five-line `mpmath` script with `mp.dps = 50`. The corresponding Lean lemmas are closed by `native_decide` (working at exact integer or rational arithmetic, not at floating-point precision) and are exhibited in their named modules in `ProjetCollatz/`.

## Appendix D. Seven theorems of the central chain

The central conditional non-existence theorem is exposed as seven distinct theorems in `ProjetCollatz/`, each tailored to a specific downstream packaging convention (six aliases of the canonical form plus the bridge lemma `sdw_from_cf`). All seven share the kernel-3 axiom profile of Appendix B; they differ only in how the three hypotheses are bundled and which sub-branch they discharge.

### D.1. Canonical form.

**`ProjetCollatz.no_nontrivial_cycle_final`**: `Phase58PorteDeuxFinal.lean`, line 339. Three `structure` hypotheses (`BakerSeparation`, `BarinaVerification`, `ProductBoundThreshold`) as explicit arguments. This is the form quoted in §3.1 and is the canonical entry point for downstream consumers.

### D.2. Variants of the canonical form.

**`ProjetCollatz.no_nontrivial_cycle_phase59`**: `Phase59ContinuedFractions.lean`, line 236. Three-argument version that takes `BakerSeparation`, `BarinaVerification`, and a continued-fraction-side hypothesis structure `DerivedLargeKBound` (with signature $\forall n, k.$ `IsOddCycle`$(n,k) \Rightarrow k > 1322 \Rightarrow n < 2^{71}$) instead of `ProductBoundThreshold`. The conversion `DerivedLargeKBound` $\rightarrow$ `ProductBoundThreshold` is performed by `sdw_from_cf` (below); the two forms are interderivable under the standard hypotheses.

**`ProjetCollatz.no_nontrivial_cycle_derived`**: `Phase56Bloc18Complete.lean`, line 355. Variant taking `ExternalCycleHypothesesDerived` (a structure that bundles the three hypotheses into a single record). Equivalent to the canonical form via record projection.

**`ProjetCollatz.no_nontrivial_cycle_full`**: `Phase52SteinerEquation.lean`, line 202. Foundation form taking `ExternalCycleHypothesesFull` (which augments the three fields with an explicit cycle-element bound below $2^{71}$). Used as the constructive seed of the alias chain; the cycle-element bound of `ExternalCycleHypothesesFull` is discharged by Barina's verification.

### D.3. Sub-branch theorems.

**`ProjetCollatz.no_cycle_k_le_1322`**: `Phase58PorteDeuxFinal.lean`, line 134. Closes the optimal sub-branch $k \leq 1322$ (Baker + Barina). Used directly by §7.1 (Reformulation 7.1, low-complexity disjunct).

**`ProjetCollatz.no_cycle_k_gt_1322`**: `Phase59ContinuedFractions.lean`, line 173. Closes the sub-branch $k > 1322$ (continued fractions + Barina), under `DerivedLargeKBound`. Used by `no_nontrivial_cycle_phase59` above as one of the two cases.

### D.4. Conditional bridge lemma.

**`ProjetCollatz.sdw_from_cf`**: `Phase59ContinuedFractions.lean`, line 197. The conditional derivation `BakerSeparation + BarinaVerification + DerivedLargeKBound` $\rightarrow$ `ProductBoundThreshold`. Witnesses the interchangeability of the two strong-hypothesis structures and is the key step in proving `no_nontrivial_cycle_phase59`.

### D.5. Logical relationships.

The seven theorems form a small commutative diagram (informally):

$$\begin{array}{c}\texttt{BakerSeparation}\\ \texttt{BarinaVerification}\\ \texttt{DerivedLargeKBound}\end{array} \xrightarrow{\texttt{sdw\_from\_cf}} \texttt{ProductBoundThreshold} \Rightarrow \texttt{no\_nontrivial\_cycle\_final}$$

with the canonical form `no_nontrivial_cycle_final` being the single endpoint, and the various intermediate variants (`derived`, `full`, `phase59`) corresponding to alternative ways to package the hypotheses into Lean structures. The two sub-branches `no_cycle_k_le_1322` and `no_cycle_k_gt_1322` are the case split that the canonical form composes.

The probe `probes/check_central_axioms.lean` runs `#print axioms` for all seven, certifying that each lives strictly under the kernel-3 baseline of Appendix B.